Ionization Energy Without Differencing:
One Constant for the Alkali Column

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Abstract

A first ionization energy is a small difference of large quantities: for potassium it is 0.027% of the total electronic energy. Any method whose totals carry a per-cent residual therefore cannot obtain it by differencing two solves — the signal falls below the error bar on each. This is why the standard formalism computes ionization by separate machinery rather than by subtracting ground-state energies, and why a parameter-free continuum model of the atom, accurate to one or two per cent on totals, is silent on ionization if asked the question that way. We report a construction that does not ask it that way. Screening the core to its net charge and giving the valence its own domain outside a radius R makes the core energy cancel *analytically* rather than numerically, leaving a problem whose energy *is* the ionization energy. For one valence electron it is exact at $R = 0$, returning hydrogen's 0.500, and it carries no partition between electrons — so the one feature of the model known to misbehave, a hard partition that over-separates two densities relative to Coulomb correlation, is absent. With no outer cutoff the radius is then the only length in the problem, and it is not free: against Shannon's six-coordinate crystal radii a single constant, $R = 0.439\, r_{\rm ion}$, reproduces the measured first ionization energies of lithium, sodium, potassium, rubidium and caesium to within 4%, four of the five to 2%, robustly to the choice of radius convention. One parameter against five observables is a claim that can fail. We state what it is not: the radius is supplied from measurement rather than computed, so this is a correlation between two measured quantities mediated by the model; gold does not obey it; and with two valence electrons the partition returns, costing 14% at $R = 0$.

1 The arithmetic that forbids differencing

Write IE $= E(\text{ion}) - E(\text{atom})$. Both terms are large and nearly equal. Table 1 gives the fraction of the total that the difference represents, together with what a 1.6% residual on the total — the mean error of the continuum model of [6] across the first three periods — amounts to in the same units.

From boron onward the quantity sought is smaller than the uncertainty on the quantities it is extracted from. At sodium it is one fourteenth of that uncertainty; at potassium one fiftieth. The conclusion does not depend on which model supplies the totals. It is arithmetic, and it applies to any method with a per-cent residual.

Two consequences follow. The first is that the standard formalism's practice is not a historical accident: Koopmans' theorem, ΔSCF, equation-of-motion coupled cluster and *GW* exist because

Table 1: A first ionization energy as a fraction of the total electronic energy it would have to be differenced out of, and the signal-to-error-bar ratio at a 1.6% residual on totals. Measured values from [1].

	$\lvert E_{\text{tot}}\rvert$ (Ha)	IE (Ha)	IE/$\lvert E_{\text{tot}}\rvert$	signal/error bar
He	2.903	0.9036	31.1%	19.4
Li	7.478	0.1981	2.65%	1.66
Be	14.667	0.3425	2.34%	1.46
B	24.654	0.3049	1.24%	0.77
C	37.845	0.4138	1.09%	0.68
N	54.589	0.5341	0.98%	0.61
O	75.067	0.5001	0.67%	0.42
F	99.734	0.6403	0.64%	0.40
Ne	128.94	0.7925	0.62%	0.38
Na	162.25	0.1886	0.116%	0.07
K	599.16	0.1595	0.027%	0.02

ionization is computed by machinery built for it, not by subtracting two ground-state energies [3, 4, 5]. The second is that a model reporting totals to one or two per cent, and declining to report ionization energies, is not thereby incomplete in a way the totals conceal — it is declining to divide a number by something smaller than its own error bar.

ΔSCF nevertheless succeeds where the core is nearly unchanged between the two solves, because then most of the error cancels between them. The construction below makes that cancellation exact instead of accidental.

2 The construction

Let the atom have V valence electrons outside a closed core of $Z - V$ electrons. Replace the core by its net charge, a kernel of charge $+V$, and let the valence densities occupy the region outside a radius R. Writing $E_{\text{val}}(n)$ for the energy of n such densities on that kernel,

$$\text{IE} \;=\; E_{\text{val}}(V-1) \;-\; E_{\text{val}}(V), \tag{1}$$

and the core energy is absent from both terms rather than subtracted between them. For $V = 1$ this reduces to $\text{IE} = -E_{\text{val}}(1)$: the ionization energy *is* the valence electron's energy, and no difference is formed at all.

The electron obeys

$$-\tfrac{1}{2}\nabla^2 u \;-\; \frac{V}{r}\,u \;=\; \varepsilon\, u, \qquad r > R, \tag{2}$$

with u vanishing at $r = R$ and free at infinity, normalised to one electron. For $V > 1$ the valence densities are non-overlapping, so (1) is a difference within one composition, taken on one kernel, with the core absent from both sides.

The check in which nothing can be adjusted, and the case it does not cover. At $R = 0$ the construction has no freedom whatever. For $V = 1$ it is hydrogen, and (2) returns -0.5 analytically. That is the anchor, and it is exact rather than fitted.

For $V = 2$ the same limit is less kind, and the reason bears on everything that follows. At $R = 0$ with a kernel of $+2$, equation (1) is the full two-electron problem: helium, with its two valence densities occupying disjoint domains. Solved to convergence that construction gives $E = -3.026$ Ha, so

$$\mathrm{IE} \;=\; E(\mathrm{He}^+) - E(\mathrm{He}) \;=\; -2.000 - (-3.026) \;=\; 1.026,$$

against a measured 0.9036 — high by 14%. The cause is known and is not the radius: two densities confined to disjoint domains avoid each other *completely and permanently*, which is more avoidance than Coulomb correlation produces, so the pair repulsion comes out some four per cent low and the state over-binds. A value of 0.903 has been reported for this case [7]; it is a value the imaginary-time relaxation passes through on its way — driving that computation to its own stopping point shows the energy rising monotonically through -2.903 at about two thousand steps and reaching -2.826 by three thousand, still rising — and not the limit of the construction.

Why the $V = 1$ column is not affected. The defect just described requires two valence densities to partition against one another. With a single valence electron there is no partner, no interface, and no over-separation: equation (2) is a one-electron problem and the only approximation in it is that the core has been replaced by its net charge at radius R. That is why the quantitative claim below is confined to the alkalis, and it is a structural reason rather than a convenient one.

The outer boundary must be removed. Earlier numerical realisations confined the valence density to an annulus $R < r < 5$ a.u. [9], while the two-valence realisation [10] limits it only by the box; the two are therefore not on the same footing. That second length does real work: solved on $R < r < \infty$ instead, the energy at $R = 1.6$ changes from 0.196 to 0.1447 Ha. Since the inner radius is the quantity of interest, the outer cutoff must go, and with it the only other length in the problem. Equation (2) on $R < r < \infty$ is well posed, and we solve it as a radial eigenvalue problem converged to 10^{-8}.

3 One constant for the column

Table 2 inverts (2) for each alkali: the radius at which the computed energy equals the measured ionization energy. The question is then whether those radii are anything, or merely five numbers fitted to five numbers.

They are not five free numbers. **The required radius is a fixed fraction of the ion's measured radius.** Fixing that fraction once at $k = 0.439$ and computing the column with no per-atom adjustment gives

$$R = 0.439\, r_{\mathrm{ion}} \;\Longrightarrow\; \frac{\mathrm{IE}_{\mathrm{comp}}}{\mathrm{IE}_{\mathrm{meas}}} = 1.04,\;\; 0.97,\;\; 1.01,\;\; 1.00,\;\; 1.02 \quad (\mathrm{Li}, \mathrm{Na}, \mathrm{K}, \mathrm{Rb}, \mathrm{Cs}). \tag{3}$$

All five within 4%; four within 2%. One constant against five observables.

Robust to the convention. Shannon tabulates two six-coordinate scales differing by exactly 14 pm: crystal radii, $r(\mathrm{O}^{2-}) = 126$ pm, and effective radii, $r(\mathrm{O}^{2-}) = 140$ pm, the latter chosen

Table 2: The radius required to reproduce each measured first ionization energy, against the ion's measured six-coordinate radius [2]. Crystal scale, $r(O^{2-}) = 126$ pm, which Shannon states corresponds more closely to the physical size of ions in a solid.

	IE (Ha)	R_{fit} (a.u.)	r_{ion} (a.u.)	$R_{\text{fit}}/r_{\text{ion}}$
Li	0.19814	0.8195	1.7008	0.482
Na	0.18886	0.9084	2.1921	0.414
K	0.15952	1.2779	2.8724	0.445
Rb	0.15351	1.3756	3.1369	0.439
Cs	0.14310	1.5689	3.4204	0.459
mean				0.448
standard deviation				5.0%

for consistency with Pauling. On the effective scale the constant becomes $k = 0.484$ and the agreement is comparable — within 8% for all five, four within 2%. Only the constant moves with the convention; the correlation does not. That is what one wants: a change of definition rescales the fitted length without destroying the relation.

4 What this is not

The radius is supplied, not computed. Equation (3) relates two measured quantities — an ionization energy and an ionic radius — through a function the model provides. Given r_{ion}, the model returns the ionization energy to a few per cent. It does not produce r_{ion}. The claim is therefore weaker than a prediction from Z alone, and stronger than a fit: a fit to five observables with five parameters asserts nothing, whereas one constant reproducing five independent measurements can fail and does not.

The model's own interface is not the screening radius. It is natural to hope the radius could be read off the atom instead of from a crystal. It cannot. A space-filling radial partition locates a core–valence interface, and that interface is a property of homogenising each shell onto a sphere; the screening radius is where the enclosed charge leaves the valence seeing a net +1. The two do not track. Taken from the interface, the radius gives lithium 0.1968 against 0.1981 but sodium only half its value, the interface having grown to 5.38 a.u. where the screening radius must be near 2.1. Across one to five core shells the interfaces come out at 1.90, 4.01, 5.38, 9.18 and 9.42 a.u. while physical core radii saturate near 2–3: a partition giving every shell a nearly fixed radial width, rather than contracting it with the charge that shell sees, misplaces every interface past the first. Lithium works because its core *is* the innermost shell, where homogenisation barely moves anything.

Gold does not obey it. Gold's measured first ionization energy, 0.339 Ha, is anomalously large, and the construction requires $R \approx 0.18$ a.u. to reproduce it — of order $0.07\, r_{\text{ion}}(Au^{+})$, not 0.44. The relation of (3) is a statement about the alkali column and is not claimed beyond it. What the construction does reproduce qualitatively is the *direction*: a valence electron outside an 18+1 ending sits at a smaller radius than one outside 8+1, hence binds more tightly, which is the observed contrast between gold at 0.339 and caesium at 0.143 [1].

An all-electron route exists and needs no radius at all. Computed in three dimensions, with the valence given angular freedom over homogenised inner shells and the same ΔSCF difference throughout, the first ionization energies of the first ten elements come out with the correct sign everywhere and a mean absolute error of 18%, lithium and sodium within 1% and neon worst at 1.51× [8]. Those figures are quoted from the source cited and have not been reproduced here; given what the paragraph above records about relaxation transients, they should be recomputed to convergence before being relied upon. That route carries no parameter and is the one to prefer where it is affordable; the construction of Section 2 is a valence-only reduction, cheaper and with a supplied radius. Computed radially instead — the valence homogenised onto a sphere — the same difference gives carbon 0.26× the measured value and the valence unbinds entirely by $Z \approx 14$. The failure is in the loss of the *valence's* angular freedom, not in the difference: carbon is 0.26× radially and 0.97× with the valence free, from the same configuration.

5 Conclusion

Ionization energy cannot be obtained by differencing total energies at per-cent accuracy, because from boron onward it is smaller than the error bar on the totals. A construction that removes the core analytically rather than numerically avoids the difficulty, is exact at the two points where it has no freedom, and — once the spurious outer cutoff is removed, leaving the inner radius as the only length — reproduces the whole alkali column from a single constant times a measured ionic radius, to four per cent. The radius comes from outside the model, and the relation does not extend to gold. Within those limits it is a one-parameter account of five independent measurements, and the kind of claim that can be refuted.

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